1 Response to the Referee: Neural Bellman Operators, R95

We thank the referee for the detailed R92 report. The controlling advisory review is the report deposited on 13 October 2026 at commit `bd2945b4b4e76b6a997dbcf7548f5fe24d3e92e6`, reviewing the R92 manuscript at `d1d1fc1f2208023a65419a616bccae0e4cdf7aa4`. We continue the descendant R94 development source at `84148cd179fe58c280a91ad281c7a81656d49788`. The R94 numerical protocol and implementation are retained without changing the primary experiment after evaluation. The present revision integrates their execution into the original paper and supplies a primitive-level proof of the continuous-state and continuous-action inequalities they require.

The paper remains **Neural Bellman Operators**, with its original author, economic technology, own-future construction, complete-policy objective, and broader applications. We have not changed the subject to a separate qualification-only paper. We have instead reorganized the article around construction, complete-policy comparison, executable continuous-risk qualification, and a matched investment study. Fifteen complete historical main-text sections move intact to the numerical appendix; the section map and preserved original entry points establish that this is relocation, not deletion.

The principal additions are Propositions and Theorems in `continuous95` and their complete proofs in `proofs95`; the six-fit, four-arm original-risk experiment in `study95`; all twelve paired intervals and paid screening diagnostics in `evidence95`; and the complete inherited 252-configuration summary. All 16,384 arm-state records meet both frozen qualification targets. All twelve simultaneous policy-cost contrasts establish equivalence within the prespecified normalized margin, but none establishes strict superiority. A separately coded rational checker validates the continuous kernels on its prescribed fixtures and checks all retained row identities. These findings are not a claim of uniquely neural superiority.

1.1 M1 / B1 / B11. A focused article without replacing NBO

We implement the requested editorial focus while retaining the original scientific question and title. The article keeps the own-future affine–ReLU construction, original-owner compilation, primitive accuracy account, complete-policy comparison, joint-domain construction, signed and centered risk results, stored-continuation resource account, and the new continuous investment application. Historical calibration, finite-menu refinements, earlier cost catalogues and deployment calculations are moved as complete, byte-identical sections to the numerical appendix. The original entry points and a per-section hash map are preserved.

The neural component is now identified by its construction, representation, attained controller and resource requirements. The certificate theorem is correctly representation-neutral: a change in parameterization cannot invalidate an otherwise valid economic comparison. This does not withdraw the NBO construction. It removes the unsupported inference that a neural representation must be uniquely responsible for every certificate improvement.

1.2 M2 / B2. Theorem-level comparison and novelty

The new comparison table distinguishes established risk Bellman recursion, residual comparison, occupation duality, support domination and feature-based learning from the additional information retained by this paper. The references include risk-averse dynamic programming and its erratum, the closely related 2025 work of Ruszczynski and Yang on feature-based risk evaluation and structured improvement, and the May 2026 preprint by Ruszczynski and Zhang on multipattern risk learning. The introduction distinguishes their episode-regret object from our deterministic

continuous-primitive, fixed-accuracy construction. The horizon proof now explicitly supplies the order and cash-invariance induction needed to carry the own-future cone argument from expectation to nested risk.

We make no priority claim for conditional-risk duality, elementary centering, convex duality, or a generic support-function test. The new continuous proposition establishes joint curvature from the original nonlinear transition, includes the state-dependent capacity multiplier, proves a global value modulus, and permits imperfect multiplier stationarity through an explicit squared residual. The continuous theorem constructs sufficient original-optimum lower and complete-policy upper inequalities without an action grid. The supplement shows exactly where these results use the investment primitives. The retained NBO construction and original-owner identity address a different construction-and-implementation obligation; neither is replaced by this application.

1.3 M3. Separate analytical, finite exact, and continuous numerical results

The article now expressly distinguishes the bounded-function conditional-risk theorem, the exact finite affine-risk occupation implementation, and the outward continuous investment implementation. The new continuous risk envelope is the full density envelope under the original uniform innovation law. Bin bounds dominate every innovation in their bins; they do not replace the original law with point masses. Finite rational risk pieces, continuous innovation-bin exclusions and an arbitrary continuous risk envelope are not described as the same computational object.

Entropic risk remains analytical. The continuous experiment validates nested AVaR at the two fixed levels, not outward exponentials and logarithms. The finite 64-date, three-state catalogue is not relabeled as a continuous high-dimensional scalability experiment.

1.4 M4 / B3 / B4 / B12. One continuous economic model and a constructed certificate

The new application uses the original investment technology, continuous state and innovation spaces, the full feasible investment interval, and genuinely nested AVaR. Its last-date training targets are paid primitive calculations with known terminal continuation, not a supplied optimal-value oracle. The complete terminal controller includes downward state acquisition, outward acceptance and rational fallback.

The primitive proof establishes joint convexity despite the nonlinear cross-state transition term. It derives a global affine lower support of the one-date value from any feasible risk density and nonnegative action multipliers, charging the stationarity residual. At the first date, a quadratic support is minimized over the entire feasible action interval. Original innovation integration uses whole-bin enclosures and explicit state-modulus allowances. The resulting policy upper cost is compared with an original-optimum lower witness, not just with the fitted continuation's own optimum.

The centered identity connects this directly constructed inequality to the general theorem, with common terms cancelled before numerical enclosure. The revision additionally reports a deterministic re-accounting of all 12,288 fitted-arm observations under supremum-norm, signed-scalar and joint local-support certificates. At the frozen residual cover, neither scalar account attains the common target on a row; the joint certificate attains it on every row. This is a post-evaluation diagnostic with unchanged policies and evidence. It is not presented as a prospectively frozen runtime contest between optimally tuned centered, uncentered and scalar-risk solvers. That stronger solver-ablation requirement remains separate.

1.5 M5 / B5. Learned and conventional representations on a common task

There are six actual fitting attempts: ReLU, additive spline and projected quadratic at each of two risk levels. They use identical labels and nominal nonlinear-feature budgets, the same nonnegative convex readout solver and a common verifier. The conventional online risk-DP arm pays for its original one-date continuation calculation. Every arm is subjected to the same own-continuation and original-optimum targets, and the same implemented terminal controller determines policy cost.

The ReLU fits have narrower whole-square residual brackets and smaller training RMSE in both cases. Their costs are not identified as strictly better. The quadratic design has effective rank six, while the ReLU and spline designs have rank 35. We disclose that nominal feature matching is not rank matching; this is not a completed rank-matched approximation-budget tournament. The previous adverse held-out ReLU proposal results remain intact in the appendix.

1.6 M6 / B9. Paid work rather than counts alone

The retained ledger separates label construction, fit attempts, whole-state residual qualification, serialization and loading, calibration, action proposals, original-law integration, all-action witnesses, complete-policy cost evaluation, screening proof and output. Successful-run process time and peak memory are recorded, along with primitive query and risk-sort counts. The measured original-value witness is included for each fitted arm; online risk-DP reuses its already paid witness.

The fitted arms process the 2,048-state workloads in approximately 11.2–12.3 seconds, compared with approximately 30.8–32.5 seconds for online risk-DP in this completed run. This is a benefit of storage in this experiment, not a uniquely neural effect. Preparation, the independent rational audit and diagnostic screening are separate paid components. Two earlier tool-interrupted starts lack recoverable complete work ledgers and are not silently omitted from the execution history. Instrumented work units are not a count of every arithmetic instruction. A complete two-architecture cost distribution, energy measurements, a monetary calibration and an exact-versus-interval crossover study are not claimed.

Screening reports 4,096 candidate bins for each arm and risk-level diagnostic, the retained and excluded counts, no-cut rows, proof time and identical tail outputs. Paying for the proof makes every displayed screened calculation slightly slower. The revision retains that result rather than presenting a positive cut count as a computational saving.

1.7 M7. The complete feasible action set

The lower certificate minimizes a globally valid quadratic minorant over the full original capacity interval. The upper certificate uses a feasible action and a dominating continuous-law risk calculation. The supplement proves the global support inequality from joint curvature and the affine capacity graph; it does not infer it from sampled action values. Refining innovation bins does not change the proposed first action. All 16,384 records meet the own-continuation target of $1/64$ and the original-optimum target of $1/16$, with a largest recorded original-optimum bound below 0.022145.

The primitive curvature result holds for every even state dimension. The primary whole-state residual implementation is two-dimensional and the economic experiment has scalar control and two decision dates. These pointwise records do not assert that the same numerical target has been uniformly checked at every initial state or every action dimension.

1.8 M8 / B8 / B10. Independent continuous-kernel validation

The new Fraction implementation reconstructs the original shock polynomial, integrates the primal AVaR threshold representation and uses a separate one-date optimizer. It does not import the production interval kernel or rational fallback. Six model identities, eighteen exact feature values, all 16,384 row identities and twelve paired-inference replays pass. Sixteen prespecified continuous-policy fixtures are checked using finer 512-bin original-law enclosures; containment in the production intervals is required, not mere overlap.

This establishes independent kernel construction, not external authorship or a second full learning stack. The largest two-control/two-innovation horizon example is not independently reimplemented in this revision. The former finite exact checker and the original vector results remain preserved. The response keeps this remaining requirement explicit rather than borrowing coverage from a smaller example.

1.9 M9. Frozen prospective evaluation

The R94 protocol corrected the earlier seed-domain collision before primary execution. Training, calibration and evaluation use domain-separated generators. The local R95 run serializes fitted models and the calibration selection before opening the evaluation stream; the evaluator verifies their hashes and frozen numerical sources. No evaluation-dependent refit, sample-size increase, representation selection or silent conventional-policy replacement occurs. Online risk-DP is selected at both risk levels, and every other arm and contrast is retained.

The cloud R94 execution had stopped before the scientific steps because its source-branch push was rejected. The local freeze was written before local evaluation but was not remotely published or externally timestamped. These provenance distinctions are recorded. The additional certificate re-accounting is explicitly post-evaluation and is not reclassified as a prespecified prospective experiment.

1.10 M10 / B6. A declared margin and separate uncertainty components

The experiment fixes 2,048 evaluation states per risk level and a normalized equivalence/noninferiority margin of 1/16, with twelve contrasts in a 95% simultaneous family. The supplement proves the paired support bound from the original action charge, value modulus and terminal allowance, and proves the outward variance bound used in the empirical-Bernstein intervals. All twelve intervals lie strictly within the equivalence margin and all include zero. Equivalence at the declared accuracy is identified; strict superiority is not.

The estimand is the initial-distribution mean of the original nested-risk policy value. It is not a sample of cash costs or a field estimate of welfare. The margin corresponds to four fifths of the maximum first-date investment charge in the normalized model; application-specific economic importance still requires calibration. Numerical integration, observation, terminal optimization and arithmetic allowances are distinguished from sampling. The declared initial grid is the target initial distribution; no claim about an uncharged continuous-initial-distribution approximation is made. Innovation bins remain numerical covers of the original continuous transition law.

1.11 M11 / B7 / B8. Horizon and dimension

The new horizon-uniform proposition supplies a fixed-original-accuracy construction for both the original scalar economy and its retained two-control, two-innovation extension under nested AVaR. It proves a uniform $27/d$ state modulus, state curvature $107/(128d)$, and action curvature $7/4$.

Explicit state, action and innovation resolutions give an original-optimum bound independent of the horizon. Stored construction then has linear-in-horizon rational-operation counts at fixed accuracy and dimension, with a separate linear-in-horizon rational bit-length bound; no constant bit cost is assumed. The proof keeps the continuous innovation law, pays for its bin approximation and reserves a separate sensor/action-rounding allowance. The original own-future construction and stored-successor resource account remain in the article, including their approximation, coverage, feasible-action and evaluation-cost conditions. This revision neither declares the general NBO program short-horizon only nor claims that the new two-date experiment removes the earlier descendant-tree bottleneck. The new curvature theorem applies to every even state dimension, while the implemented primary state cover is two-dimensional. The original largest-vector horizon table and its unfavorable scaling remain in the appendix as a distinct benchmark obligation.

1.12 M12 / B11. Preservation and editorial compression

Fifteen complete main-text sections are relocated to the numerical appendix. Their hashes agree exactly with the reviewed R92 sources. The original wrappers are preserved, and the content map locates every former active section in exactly one current document. The main text retains the NBO construction and complete-policy theory and uses one new continuous economic model to connect the central implementation claims. No unfavorable observation, earlier theorem, economic primitive, or broader application is discarded. The final source-bound document manifest reports actual compiled page counts; historical PDFs are not renamed as current PDFs.

1.13 Minor and technical comments

1. The centered penalty retains infimum zero without asserting attainment; finite maximizing pieces and continuum densities are distinguished.
2. Occupations in the risk result remain risk-distorted, not physical or empirical occupation probabilities.
3. The continuous theorem now has one explicit assumption paragraph, with bounded domains, feasible risk densities, compact action graphs, selection and outward arithmetic conditions collected together.
4. All 252 inherited interval configurations are summarized. Every configuration has a strictly tighter state-dependent bound; there are zero equal or looser cases and 118 no-cut configurations. The complete distribution, not only a maximum, is retained.
5. The finite catalogue has 91,260 candidate pieces and 73,549 retained pieces across its configurations. The continuous screen has its own 4,096-bin denominators. These are different objects, and their proof costs are not conflated.
6. The existing rational bit-growth evidence is retained. No new per-operation arithmetic crossover is reported without measurement.
7. The absorbing example is explicitly zero policy loss with nonzero continuation bias.
8. Strict screening retains equality. Invalid intervals and equality cases are tested in both the retained and new checks.
9. Entropic statements remain analytical; the AVaR calculation is not cited as an outward exponential/logarithm implementation.
10. Every claim about the new lower certificate refers to the full original action interval. A finite menu or sampled grid is not called an all-action certificate.
11. The joint-domain and population theorems are not used as empirical generalization evidence. The whole-state residual cover is a deterministic inequality, and independent observations

are counted separately.

12. Detailed source and release records are placed in the reproducibility materials rather than repeated as the central scientific argument.

1.14 Verification and delivery status

The completed local continuous experiment and independent kernel audit have source-bound records. The new 34-test continuous suite and 25-test R95 suite pass without skips. The complete inherited suite is attempted separately. The downloaded publication artifact intentionally excludes older code and datasets, and the resulting missing-fixture failures remain recorded. They are not evidence of a new mathematical failure, but they also do not constitute a passed full-repository replay.

The publication continuation retains the original local receipts and their limitations as historical records. It restores the inherited repository dependencies, publishes the revision through new R95 branches, and separates the original local experiment from replay of the same frozen models and initial-state streams. Current repository replay, source-bound typesetting and remote delivery are recorded separately under audit/remote95 and RELEASE95.json. A later execution of the same stream is a replication, not an increase in independent observations; an earlier local freeze is not retrospectively relabelled a remote preregistration. The source text and current release receipts, rather than inherited status sentences, identify the verification actually completed.